

Question description

A long time ago, there was a desolate village in India. The ancient buildings, streets, and businesses were deserted. The windows were open, and the stairwell was in disarray. You can be sure that it will be a fantastic area for mice to romp about in! People in the community have now chosen to provide high-quality education to young people in order to further the village's growth.

As a result, they established a programming language coaching centre. People from the coaching centre are presently performing a test. Create a programme for the `GetNth()` function, which accepts a linked list and an integer index and returns the data value contained in the node at that index position.

Example

Input: 1->10->30->14, index = 2

Output: 30

The node at index 2 is 30

Question No:6

Constraints

$1 < N < 1000$

$1 < X < 1000$

$1 < I < 1000$

Input Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third Line Index I

Output Format

First Line indicates the linked list

second line indicates the node at indexing position

Logical Test Cases

Test Case 1

INPUT (STDIN)

4
1 2 3 4
2

EXPECTED OUTPUT

Linked list:-->4-->3-->2-->1
Node at index=2:3

Test Case 2

INPUT (STDIN)

4
11 21 31 41
3

EXPECTED OUTPUT

Linked list:-->41-->31-->21-->11
Node at index=3:21

Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

struct node *next;

Test Case 3

KEYWORD

int GetNth(struct node*
head,int index)

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct Node {
5      int data;
6      struct Node *next;
7  };
8
9  struct Node* createNode(int data) {
10     struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
11
12     newNode->data = data;
13     newNode->next = NULL;
14
15     return newNode;
16 }
17
18 void insertAtBeginning(struct Node **head, int data) {
19     struct Node *newNode = createNode(data);
20
21     newNode->next = *head;
22     *head = newNode;
23 }
24
25 int getNode(struct Node *head, int index) {
26     struct Node *temp = head;
27     int count = 1;
```

```
29     while (temp != NULL) {
30         if (count == index) {
31             return temp->data;
32         }
33
34         count++;
35         temp = temp->next;
36     }
37
38     return -1;
39 }
40
41 void display(struct Node *head) {
42     struct Node *temp = head;
43
44     printf("Linked list:");
45
46     while (temp != NULL) {
47         printf("-->%d", temp->data);
48         temp = temp->next;
49     }
50
51     printf("\n");
52 }
```

```

54  int main() {
55      int n, index;
56      struct Node *head = NULL;
57
58      scanf("%d", &n);
59
60      for (int i = 0; i < n; i++) {
61          int value;
62          scanf("%d", &value);
63
64          insertAtBeginning(&head, value);
65      }
66
67      scanf("%d", &index);
68
69      display(head);
70
71      printf("Node at index=%d:%d", index, getNode(head, index));
72
73      return 0;
74  }

```

Output:

Test Case-1:

```

4
1 2 3 4
2
Linked list:-->4-->3-->2-->1
Node at index=2:3

```

Test Case-2:

```
4
11 21 31 41
3
Linked list:-->41-->31-->21-->11
Node at index=3:21
PS C:\Users\kesha\OneDrive\Documents\3sem\C> █
```

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